

**NAME: MOHAMMAD UBAIDULLAH**  
**ROLL NO : 2403A510B9**  
**BATCH NO :24BTCAICSB04**  
**COURSE NAME : AI ASSISTED CODING**  
**COURSE CODE : 24CS002PC215**

|                                                                                         |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
|-----------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------------------|----------------------|-------------------|------------------------|------------------|-------------------------|-------------------|--------------------|--------------------|-------------------|----------------|---------------|------------|------------------------|-----------------------|-------------------|-----------------|
| <b>SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE</b>                           |                  | <b>DEPARTMENT OF COMPUTER SCIENCE ENGINEERING</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| <b>ProgramName:</b> B. Tech                                                             |                  | <b>Assignment Type:</b> Lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>AcademicYear:</b> 2025-2026 |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| <b>CourseCoordinatorName</b>                                                            |                  | Venkataramana Veeramsetty                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| <b>Instructor(s)Name</b>                                                                |                  | <table border="1"> <tr><td>Dr. V. Venkataramana (Co-ordinator)</td></tr> <tr><td>Dr. T. Sampath Kumar</td></tr> <tr><td>Dr. Pramoda Patro</td></tr> <tr><td>Dr. Brij Kishor Tiwari</td></tr> <tr><td>Dr.J.Ravichander</td></tr> <tr><td>Dr. Mohammand Ali Shaik</td></tr> <tr><td>Dr. Anirodh Kumar</td></tr> <tr><td>Mr. S.Naresh Kumar</td></tr> <tr><td>Dr. RAJESH VELPULA</td></tr> <tr><td>Mr. Kundhan Kumar</td></tr> <tr><td>Ms. Ch.Rajitha</td></tr> <tr><td>Mr. M Prakash</td></tr> <tr><td>Mr. B.Raju</td></tr> <tr><td>Intern 1 (Dharma teja)</td></tr> <tr><td>Intern 2 (Sai Prasad)</td></tr> <tr><td>Intern 3 (Sowmya)</td></tr> <tr><td>NS_2 ( Mounika)</td></tr> </table> |                                | Dr. V. Venkataramana (Co-ordinator) | Dr. T. Sampath Kumar | Dr. Pramoda Patro | Dr. Brij Kishor Tiwari | Dr.J.Ravichander | Dr. Mohammand Ali Shaik | Dr. Anirodh Kumar | Mr. S.Naresh Kumar | Dr. RAJESH VELPULA | Mr. Kundhan Kumar | Ms. Ch.Rajitha | Mr. M Prakash | Mr. B.Raju | Intern 1 (Dharma teja) | Intern 2 (Sai Prasad) | Intern 3 (Sowmya) | NS_2 ( Mounika) |
| Dr. V. Venkataramana (Co-ordinator)                                                     |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| Dr. T. Sampath Kumar                                                                    |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| Dr. Pramoda Patro                                                                       |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| Dr. Brij Kishor Tiwari                                                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| Dr.J.Ravichander                                                                        |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| Dr. Mohammand Ali Shaik                                                                 |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| Dr. Anirodh Kumar                                                                       |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| Mr. S.Naresh Kumar                                                                      |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| Dr. RAJESH VELPULA                                                                      |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| Mr. Kundhan Kumar                                                                       |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| Ms. Ch.Rajitha                                                                          |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| Mr. M Prakash                                                                           |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| Mr. B.Raju                                                                              |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| Intern 1 (Dharma teja)                                                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| Intern 2 (Sai Prasad)                                                                   |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| Intern 3 (Sowmya)                                                                       |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| NS_2 ( Mounika)                                                                         |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| <b>CourseCode</b>                                                                       | 24CS002PC215     | <b>CourseTitle</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | AI Assisted Coding             |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| <b>Year/Sem</b>                                                                         | II/I             | <b>Regulation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | R24                            |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| <b>Date and Day of Assignment</b>                                                       | Week1 - Thursday | <b>Time(s)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| <b>Duration</b>                                                                         | 2 Hours          | <b>Applicable to Batches</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 24CSBTB01 To 24CSBTB39         |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| <b>AssignmentNumber:</b> 2.4(Present assignment number)/24(Total number of assignments) |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
|                                                                                         |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |
| <b>Q.No.</b>                                                                            | <b>Question</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>ExpectedTime</b>            |                                     |                      |                   |                        |                  |                         |                   |                    |                    |                   |                |               |            |                        |                       |                   |                 |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                        |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>to<br/>complete</b> |
| 1 | <p>Lab 2: Exploring Additional AI Coding Tools – Gemini (Colab) and Cursor AI</p> <p><b>Lab Objectives:</b></p> <ul style="list-style-type: none"> <li>• To explore and evaluate the functionality of Google Gemini for AI-assisted coding within Google Colab.</li> <li>• To understand and use Cursor AI for code generation, explanation, and refactoring.</li> <li>• To compare outputs and usability between Gemini, GitHub Copilot, and Cursor AI.</li> <li>• To perform code optimization and documentation using AI tools.</li> </ul> <p><b>Lab Outcomes (LOs):</b><br/>After completing this lab, students will be able to:</p> <ul style="list-style-type: none"> <li>• Generate Python code using Google Gemini in Google Colab.</li> <li>• Analyze the effectiveness of code explanations and suggestions by Gemini.</li> <li>• Set up and use Cursor AI for AI-powered coding assistance.</li> <li>• Evaluate and refactor code using Cursor AI features.</li> <li>• Compare AI tool behavior and code quality across different platforms.</li> </ul> <p><b>Task Description #1</b></p> <ul style="list-style-type: none"> <li>• Open Google Colab and use Google Gemini to generate Python code that performs sorting of a list using both the bubble sort algorithm and Python's built-in sort() function. Compare the two implementations.</li> </ul> <p><b>Expected Output #1</b></p> <ul style="list-style-type: none"> <li>• Two sorting implementations: Bubble sort (manual logic) and Built-in sort()</li> </ul> <p><b>TASK1:</b></p> <p><b>PROMPT 1:</b></p> <p>generate Python code that performs sorting of a list using both the bubble sort algorithm and Python's built-in sort() function. Compare the two implementations.</p> | Week1 -<br>Thursday    |

```

# Simple sorting using Python's built-in function
def python_sorted(arr):
    return sorted(arr)

my_list = [5, 2, 8, 1, 9, 4]
sorted_list = python_sorted(my_list)
print("Original list:", my_list)
print("Sorted list (using sorted()):", sorted_list)

# Bubble Sort implementation
def bubble_sort(arr):
    n = len(arr)
    # Create a copy to avoid modifying the original list
    arr_copy = arr[:]
    for i in range(n):
        # Last i elements are already in place
        for j in range(0, n - i - 1):
            # Traverse the array from 0 to n-i-1
            # Swap if the element found is greater than the next element
            if arr_copy[j] > arr_copy[j + 1]:
                arr_copy[j], arr_copy[j + 1] = arr_copy[j + 1], arr_copy[j]
    return arr_copy

my_list_for_bubble_sort = [64, 34, 25, 12, 22, 11, 90]
bubble_sorted_list = bubble_sort(my_list_for_bubble_sort)
print("\nOriginal list for Bubble Sort:", my_list_for_bubble_sort)
print("Sorted list (using Bubble Sort):", bubble_sorted_list)

```

Original list: [5, 28, 78, 1, 9, 4]  
Sorted list (using sorted()): [1, 4, 5, 9, 28, 78]

Original list for Bubble Sort: [64, 34, 25, 12, 22, 11, 90]  
Sorted list (using Bubble Sort): [11, 12, 22, 25, 34, 64, 90]

### Task Description #2

- In Colab, use Google Gemini to generate a Python function that takes a string and returns: The number of vowels, The number of consonants, The number of digits in the string

### Expected Output #2-

- Complete function that Iterates through characters of a string and Counts vowels, consonants, and digits

## Task 2:

### Prompt 2:

take a string which contains of digits and alphabets .Then write a python code to count the numbers of digits and constants and vowels can you take string in dyanamic method

```

def count_chars(input_string):
    digits = 0
    vowels = 0
    consonants = 0
    input_string = input_string.lower() # Convert to lowercase for easier vowel counting

    for char in input_string:
        if char.isdigit():
            digits += 1
        elif char.isalpha():
            if char in 'aeiou':
                vowels += 1
            else:
                consonants += 1
    return digits, vowels, consonants

# Example usage:
my_string = input("Please enter a string:") # Get input dynamically from the user
num_digits, num_vowels, num_consonants = count_chars(my_string)

print(f"The string '{my_string}' contains:")
print(f"Digits: {num_digits}")
print(f"Vowels: {num_vowels}")
print(f"Consonants: {num_consonants}")

```

```

➔ Please enter a string:AHUEYDWGU1266213agdfdsdfgscb
The string 'AHUEYDWGU1266213agdfdsdfgscb' contains:
Digits: 7
Vowels: 5
Consonants: 16

```

### Task Description #3

- Install and set up Cursor AI. Use it to generate a Python program that performs file handling:
- Create a text file
- Write sample text
- Read and display the content

### Expected Output #3

- Functional code that creates a .txt file, writes content to it, and reads it back.
- Screenshot of Cursor AI interface showing: Prompt used, Generated code, Output of file operations

## Task 3:

### Prompt 3:

Write a python programm the Create a file and write a text in the file and display the content of the file by using file handling

```
[3] # Define the filename
filename = "my_example_file.txt"

# Text to be written to the file
file_content = "This is some sample text that will be written to the file."
file_content += "This is the second line."

# Create and write to the file
try:
    with open(filename, "w") as file:
        file.write(file_content)
    print(f"Successfully created and wrote to '{filename}'")

except IOError as e:
    print(f"Error writing to file: {e}")

# Read and display the content of the file
try:
    with open(filename, "r") as file:
        read_content = file.read()
    print(f"\nContent of '{filename}':")
    print(read_content)

except IOError as e:
    print(f"Error reading file: {e}")
```

```
⇒ Successfully created and wrote to 'my_example_file.txt'

Content of 'my_example_file.txt':
This is some sample text that will be written to the file.
This is the second line.
```

#### Task Description #4

- Ask Google Gemini to generate a Python program that implements a simple calculator using functions (add, subtract, multiply, divide). Then, ask Gemini to explain how the code works.

#### Expected Output #4

- Complete calculator code with user input and operation selection.
- Line-by-line explanation or markdown-style explanation provided by Gemini.
- Screenshot of both the code and explanation in Colab.

Task4:

Prompt 4:

write a python program that implements a simple calculator. Modify the code and add the comments. Explain iterations and explain the code by line by line.

```
def add(x, y):
    """Adds two numbers."""
    return x + y

def subtract(x, y):
    """Subtracts two numbers."""
    return x - y

def multiply(x, y):
    """Multiplies two numbers."""
    return x * y

def divide(x, y):
    """Divides two numbers."""
    if y == 0:
        return "Error: Cannot divide by zero"
    return x / y

# Get input from the user
try:
    num1 = float(input("Enter first number: ")) # Prompt user for the first number and convert to float
    num2 = float(input("Enter second number: ")) # Prompt user for the second number and convert to float

    print("Select operation:")
    print("1. Add")
    print("2. Subtract")
    print("3. Multiply")
    print("4. Divide")

    choice = input("Enter choice(1/2/3/4): ") # Prompt user for their choice of operation

    if choice == '1': # Check if the choice is addition
        print(f"{num1} + {num2} = {add(num1, num2)}") # Perform addition and print the result
    elif choice == '2': # Check if the choice is subtraction
        print(f"{num1} - {num2} = {subtract(num1, num2)}") # Perform subtraction and print the result
    elif choice == '3': # Check if the choice is multiplication
        print(f"{num1} * {num2} = {multiply(num1, num2)}") # Perform multiplication and print the result
    elif choice == '4': # Check if the choice is division
        print(f"{num1} / {num2} = {divide(num1, num2)}") # Perform division and print the result
    else:
        print("Invalid input") # Print error for invalid input

except ValueError:
    print("Invalid input. Please enter numbers.") # Print error if input is not a valid number
```

```
Enter first number: 78
Enter second number: 9
Select operation:
1. Add
2. Subtract
3. Multiply
4. Divide
Enter choice(1/2/3/4): 2
78.0 - 9.0 = 69.0
```

### Task Description #5

- Use Cursor AI to create a Python program that checks if a given year is a leap year or not. Try different prompt styles and see how Cursor modifies its code suggestions.

### Expected Output #5

- A functional program to check leap year with sample input/output
- At least two versions of the code (from different prompts)
- A short comparison of which version is better and why

## Task5:

## Prompt5:

genrate a python programme that checks whether the give year or not

```
def is_leap_year_v1(year):  
    """  
    Checks if a year is a leap year using nested if-else statements.  
    """  
    if (year % 4) == 0:  
        if (year % 100) == 0:  
            if (year % 400) == 0:  
                return True  
            else:  
                return False  
        else:  
            return True  
    else:  
        return False  
  
# Test cases  
print(f"2000 is a leap year (v1): {is_leap_year_v1(2000)}")  
print(f"1900 is a leap year (v1): {is_leap_year_v1(1900)}")  
print(f"2024 is a leap year (v1): {is_leap_year_v1(2024)}")  
print(f"2023 is a leap year (v1): {is_leap_year_v1(2023)}")
```

**Note:** Report should be submitted a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots

### Evaluation Criteria:

| Criteria                                                                             | Max Marks |
|--------------------------------------------------------------------------------------|-----------|
| Two sorting implementations: Bubble sort (manual logic) and Built-in sort() (Task#1) | 0.5       |
| Counts vowels, consonants, and digits(Task#2)                                        | 0.5       |

|  |                                                                                                        |                  |  |  |
|--|--------------------------------------------------------------------------------------------------------|------------------|--|--|
|  | Functional code that creates a .txt file, writes content to it, and reads it back- Use cursor (Task#3) | 0.5              |  |  |
|  | Complete calculator code with user input and operation selection. (Task#4)                             | 0.5              |  |  |
|  | A functional program to check leap year with sample input/output-use Cursor (Task#5)                   | 0.5              |  |  |
|  | <b>Total</b>                                                                                           | <b>2.5 Marks</b> |  |  |